

Task: Cleaning k Shoes ($k = 2$)

Coarse instruction ($w(\mathcal{R}) = 3$) Finer instruction ($w(\mathcal{R}) = 2$) r_1

If the number of dirty shoes is greater than zero, then clean one shoe

 r_{1a}

r_{1a} Pick up the rag

$$r_{1a} : \{\neg H_r\} \mapsto \{H_r\}$$
 r_{1b}

r_{1b} If you're holding an unsoaked rag, soak it.

$$r_{1b} : \{H_r, \neg S_r\} \mapsto \{S_r, H_r?\}$$
 $\mathcal{R}_1$

r_{1c} With a soaked rag and dirty shoes, clean the shoes.

$$r_{1c} : \{N_f > 0, S_r\} \mapsto \{N_f \downarrow\}$$
 r_2

Once the shoes are clean,
place the rag on the floor.

$$r_2 : \{N_f = 0, \neg onfloor_r\} \mapsto \{onfloor_r\}$$
